Emerging Tech Training for AI

Courses

VM Required:

- https://vam.udemy.com/course/generative-ai-for-beginners-b/
- https://vam.udemy.com/course/intro-to-chatgpt-and-generative-ai/
- https://vam.udemy.com/course/chatgpt-a-master-guide-for-professionals

Emerging Tech

- https://www.learnpython.org/.
- Ultimate Pandas and Python Data Analysis (Complete Course) | Udemy Business
- The Complete Machine Learning Course with Python | Udemy Business
- A deep understanding of deep learning (with Python intro) | Udemy Business

Specific Follow-ups:

- LangChain- Develop LLM powered applications with LangChain | Udemy Business
- W&B 101: ML Experiment Tracking Course | Weights & Biases (wandb.courses)
- Training LLMs Course: Discover Fine-Tuning Techniques (wandb.courses)
- Building an LLM Application - LlamaIndex

Tools:

- VS Code
- AWS Account
- Azure Account
- Google Colab account
- Free OpenAI account
- WSL
    - Anaconda
    - Python, etc

General Learning Flow

Week 1: Python and Basic AI Concepts

Day 1-2: Python Fundamentals

- **Objective:** Ensure they are comfortable with Python syntax and basic programming concepts.
- **Tasks:**
  - Install Python and set up a development environment (VSCode, WSL, Anaconda, Python 3.10).
  - Cover basics: variables, data types, loops, conditionals, functions.
  - Extra Training material:
    - https://learn-python.adamemery.dev/

Day 3-4: Python for Data Science

- **Objective:** Introduce them to Python libraries commonly used in AI.
- **Tasks:**
  - Overview of libraries: NumPy, pandas, Matplotlib.
  - Practice: Data manipulation with pandas, basic plotting with Matplotlib.
  - Extra Training material:
    - 10 minutes to pandas — pandas 2.2.2 documentation (pydata.org),

Day 5: Introduction to AI and ML

- **Objective:** Familiarize them with key AI and ML concepts.
- **Tasks:**
  - Study the basics of machine learning: supervised vs. unsupervised learning, models, and algorithms.
  - Training:
    - Machine Learning for Absolute Beginners - Level 1 | Udemy Business

Week 2: Deep Learning and PyTorch

Day 6-7: Introduction to Deep Learning

- **Objective:** Understand the basics of deep learning and neural networks.
- **Tasks:**
  - Concepts: Neurons, activation functions, layers, and forward/backward propagation.
  - Study resources: Coursera or Fast.ai introductory courses.

    - Assignment: Implement a simple neural network using Python (without any deep learning library).

Day 8-9: Getting Started with PyTorch

- **Objective:** Familiarize them with PyTorch and basic operations.
- **Tasks:**
    - Install PyTorch and set up the environment.
    - Introduction to tensors and basic tensor operations.
    - Practice: Simple exercises with PyTorch tensors.

Day 10: Implementing a CNN in PyTorch

- **Objective:** Build their first Convolutional Neural Network (CNN).
- **Tasks:**
    - Study the theory behind CNNs: convolutional layers, pooling, and fully connected layers.
    - Implement a basic CNN using PyTorch on a simple dataset like MNIST.
    - Use Weights and Biases (wandb) to track experiments.

Week 3: RAG, Agents, and Advanced PyTorch

Day 11-12: Retrieval-Augmented Generation (RAG)

- **Objective:** Understand the concept and implementation of RAG.
- **Tasks:**
    - Study the basics of RAG: what it is and why it's important.
    - Explore existing implementations and walkthroughs (e.g., Hugging Face RAG implementation).
    - Assignment: Implement a simple RAG model using PyTorch.

Day 13-14: Introduction to Agents

- **Objective:** Introduce the concept of AI agents and how they can be used in AI workflows.
- **Tasks:**
    - Learn the basics of agents in AI (e.g., what they are, how they work).
    - Study an example of agent-based modeling in Python.
    - Practice: Build a basic agent that interacts with an environment.

Day 15: Advanced PyTorch Techniques

- **Objective:** Dive deeper into PyTorch for more complex tasks.
- **Tasks:**
    - Learn about advanced topics like custom layers, loss functions, and optimizers.

- Practice: Modify the CNN from the previous week to include these custom elements.
- Use Weights and Biases to experiment with different configurations.